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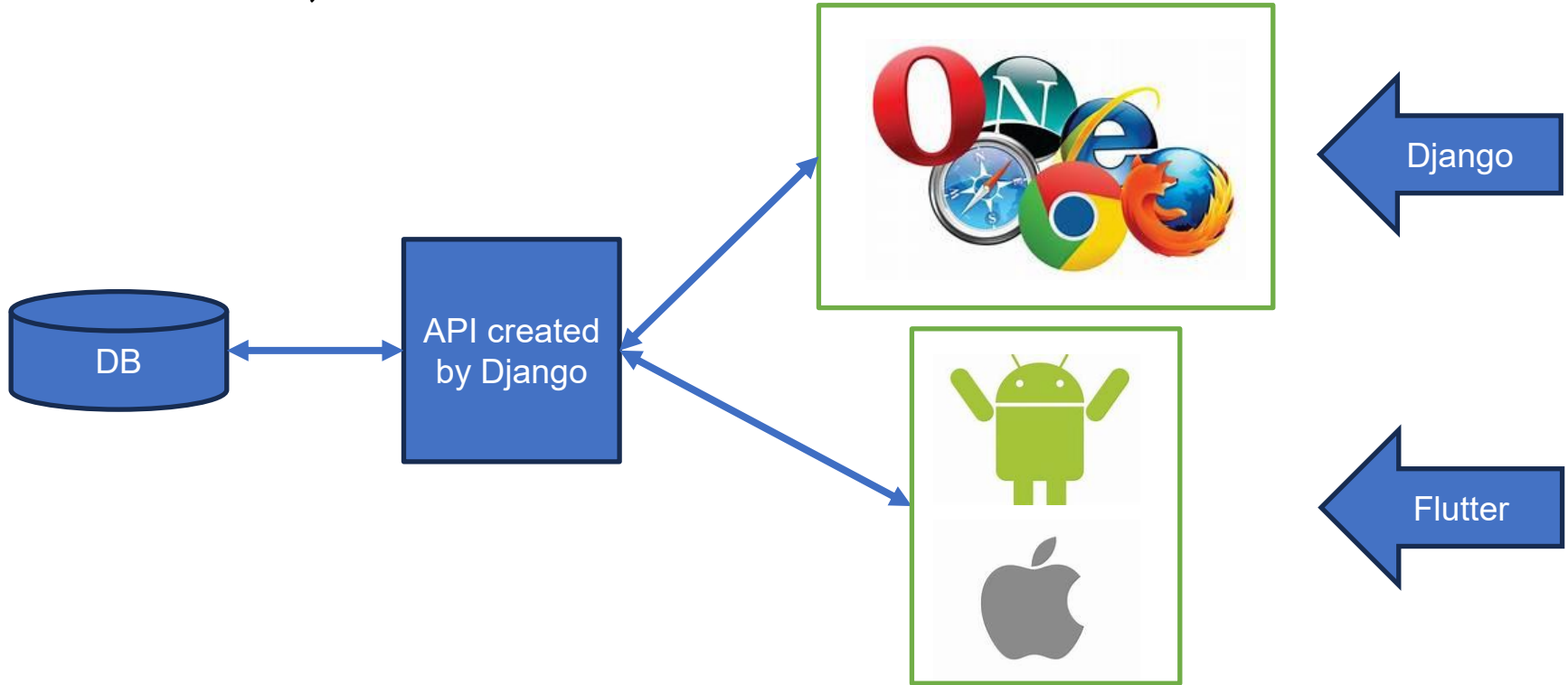
Introduction to the Internet and Web Framework

Tim Dosen PBP

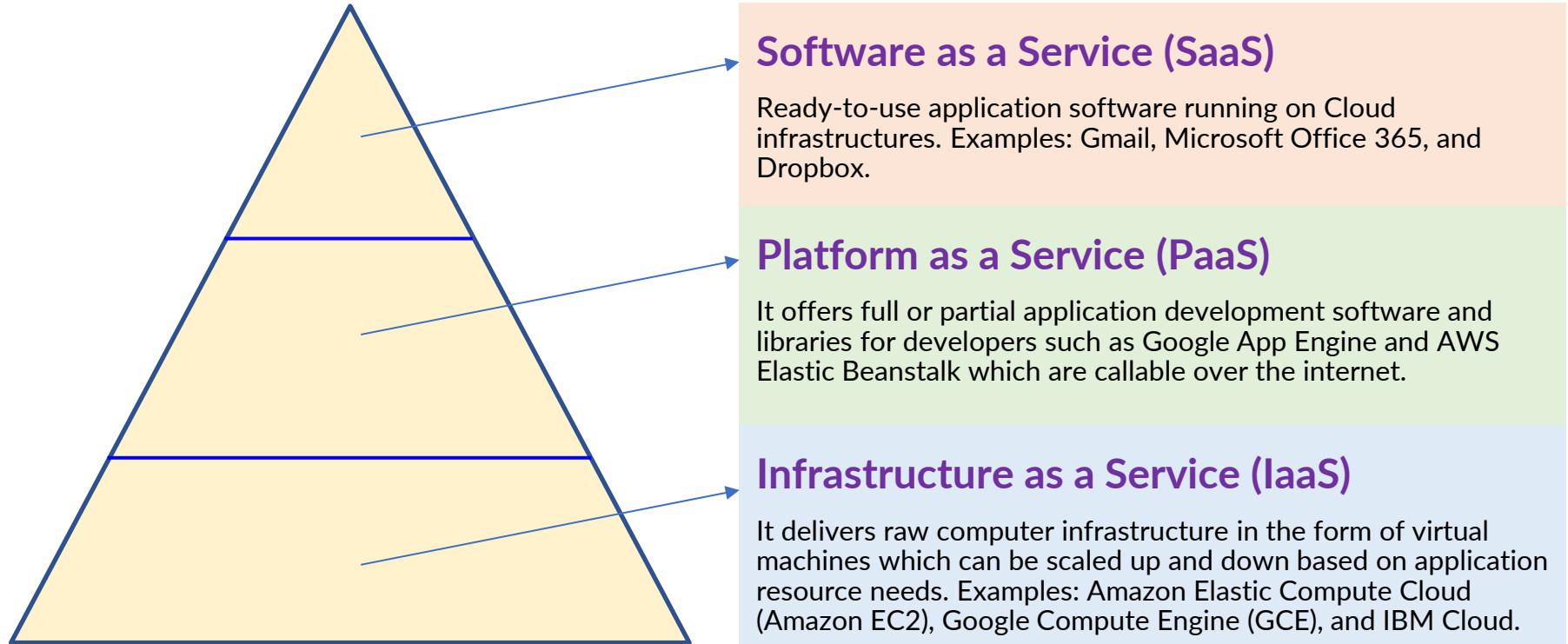
Previous Session Topics

- Basic Concepts of Platforms
- Programming Frameworks
- Web Platforms vs. Mobile Platforms

In General, we can do:



Forms of Cloud Computing



CI/CD

- **Continuous integration (CI)** automatically **builds, tests, and integrates** code changes within a shared repository
- **Continuous delivery (CD)** automatically **delivers** code changes to production-ready environments for approval
- **Continuous deployment (CD)** automatically **deploys** code changes to customers directly

Agenda

- Basic concepts of the Internet
- How webserver works
- Web framework

Introduction to the Internet

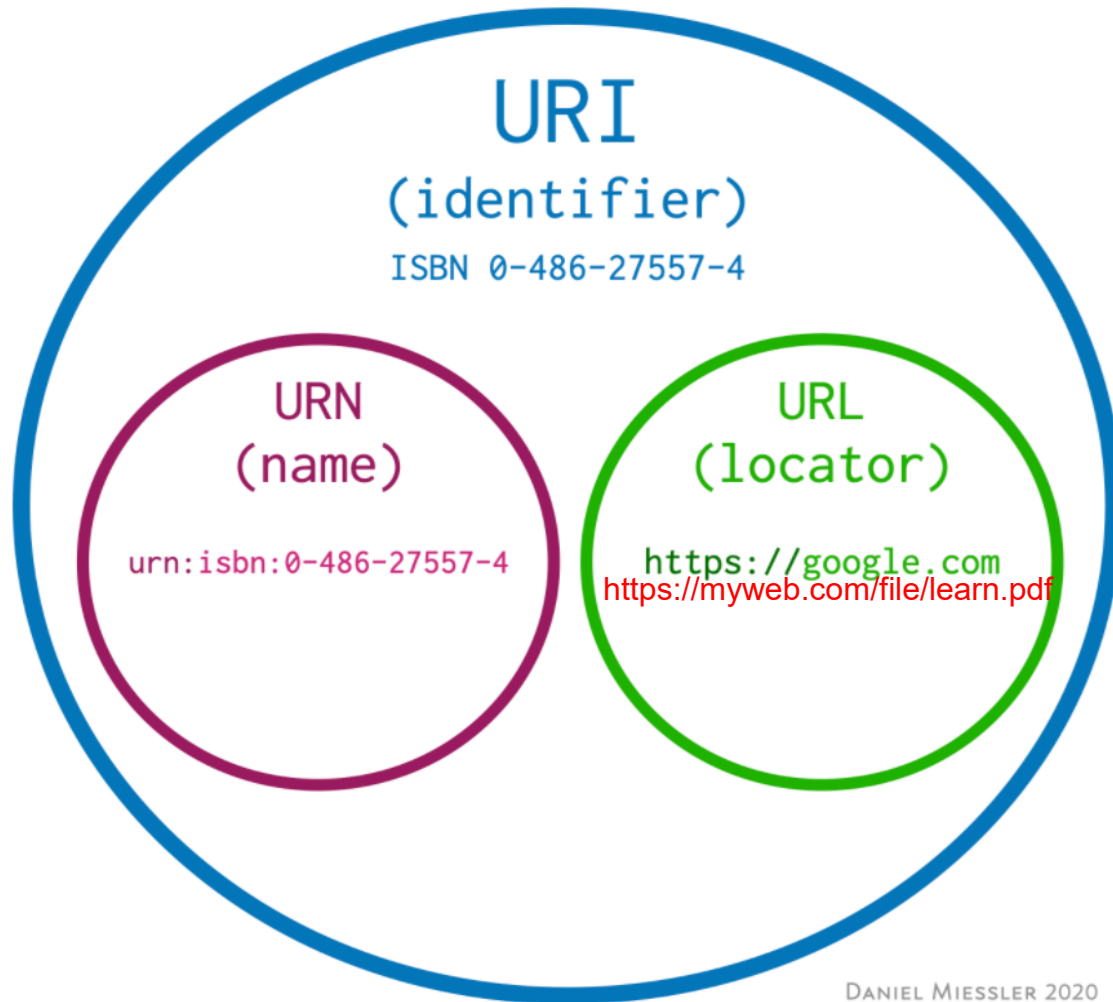
The Internet & the Web

- The Internet: a global network of computers

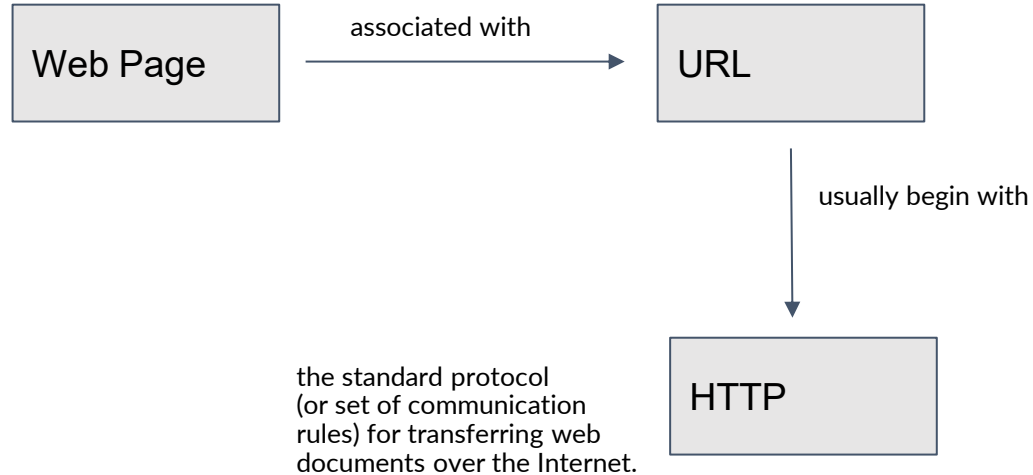


The World Wide Web (WWW, or simply Web) is an **information space** in which the items of interest, referred to as resources, are identified by global identifiers called Uniform Resource Identifiers (URI).

-(W3C's [Architecture of the World Wide Web, Volume I](#))



URL



The URL specifies the address (i.e., location) of the web page displayed in the browser window.

URL Syntax

- Every HTTP URL conforms to the syntax of a generic URI.
- The generic URI syntax consists of a hierarchical sequence of components referred to as the scheme, authority, path, query, and fragment.

URI = scheme ":" hier-part ["?" query] ["#" fragment]

hier-part = "/" authority path-abempty begins with "/" or is empty
/ path-absolute
/ path-rootless
/ path-empty

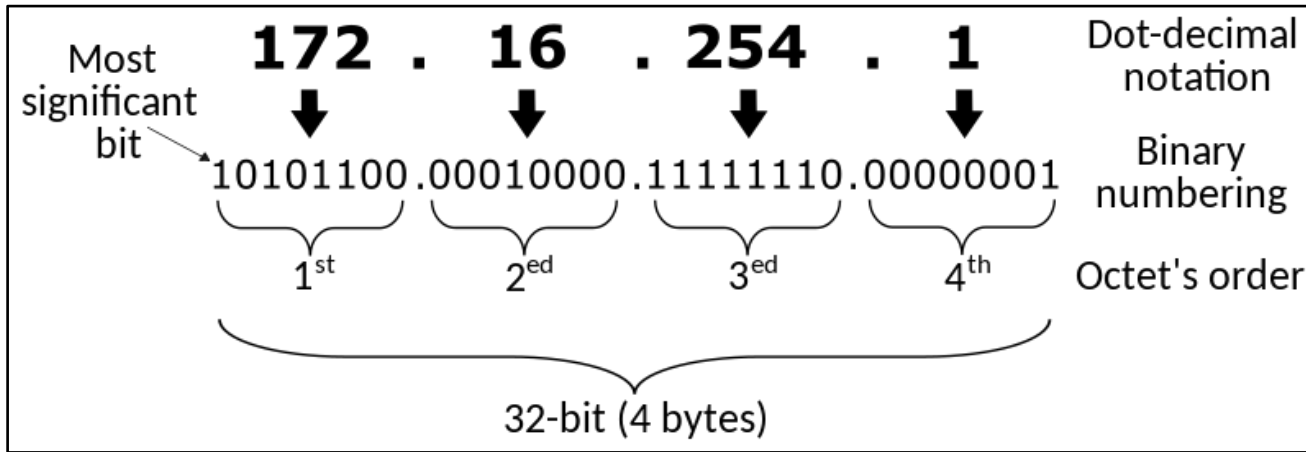
- Example: 

URL Syntax

- Example:  `http://localhost:8000/contact/person?name=pakbepe#phone`
- Examples of popular schemes include **http** (hypertext transfer protocol), **https** (hypertext transfer protocol secure), **ftp** (file transfer protocol), and **mailto**.
- The host subcomponent of authority is identified by an IP literal encapsulated within square brackets (IPv6 or IPvFuture), an IPv4 address in dotted-decimal form, or a registered name (hostname).
 - `http://localhost:8000/` → hostname
 - `http://127.0.0.1:8000/` → IPv4 address
- Hostname is translated into IP (Internet Protocol) address by DNS (Domain Name System).
- IP address determines the location of the server.
- Port number is associated with service or program on the server.

URL – IPv4 Address

- IPv4 addresses are usually represented in dot-decimal notation, consisting of four decimal numbers, each ranging from 0 to 255, separated by dots.
- Each part represents a group of 8 bits (an octet) of the address.

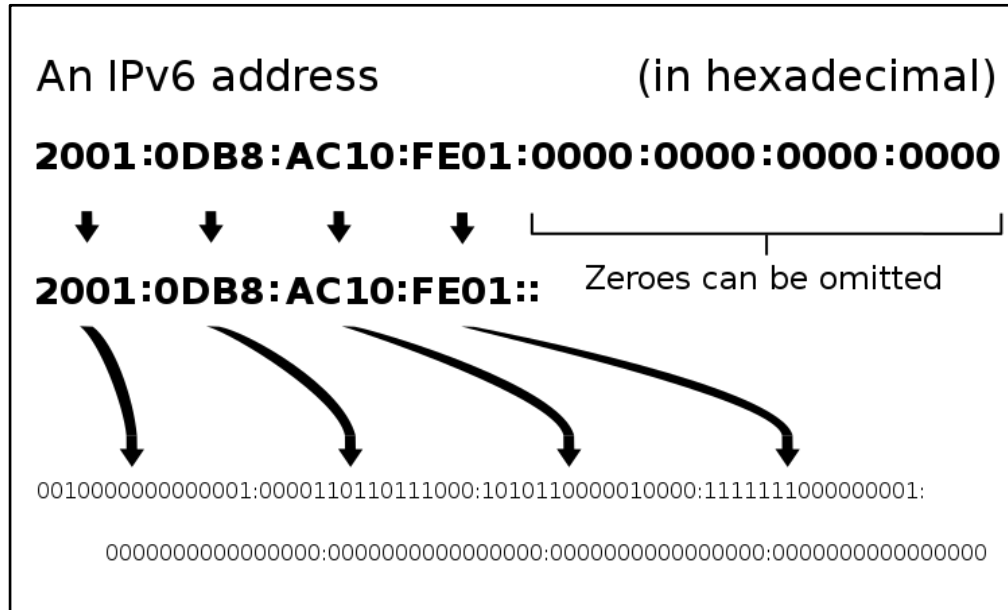


Picture from:

https://upload.wikimedia.org/wikipedia/commons/thumb/6/66/IPv4_address_structure_and_writing_systems-en.svg/799px-IPv4_address_structure_and_writing_systems-en.svg.png

URL – IPv6 Address

- In IPv6, the address size was increased from 32 bits in IPv4 to 128 bits, thus providing up to 2^{128} addresses.



URL Examples

1. <https://www.example.com/>
2. <https://www.example.com/index.html>
3. <https://www.example.com/index.php>
4. <https://example.com/>

1. <https://example.com/username/>
2. <https://example.com/username/gallery/>
3. <https://username.example.com/>
4. <https://username.example.com/gallery/>

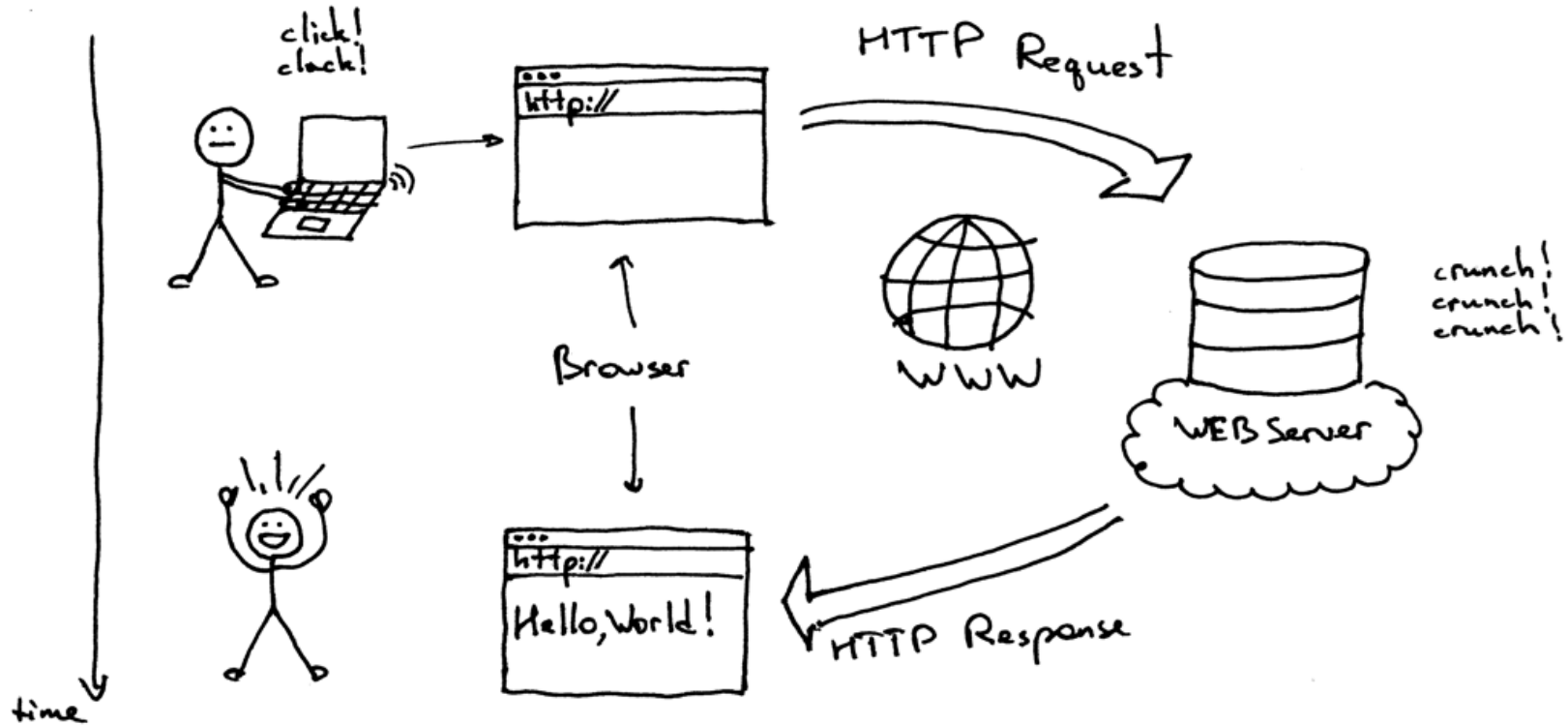
Which one
to choose?

<https://example.com/#contact>

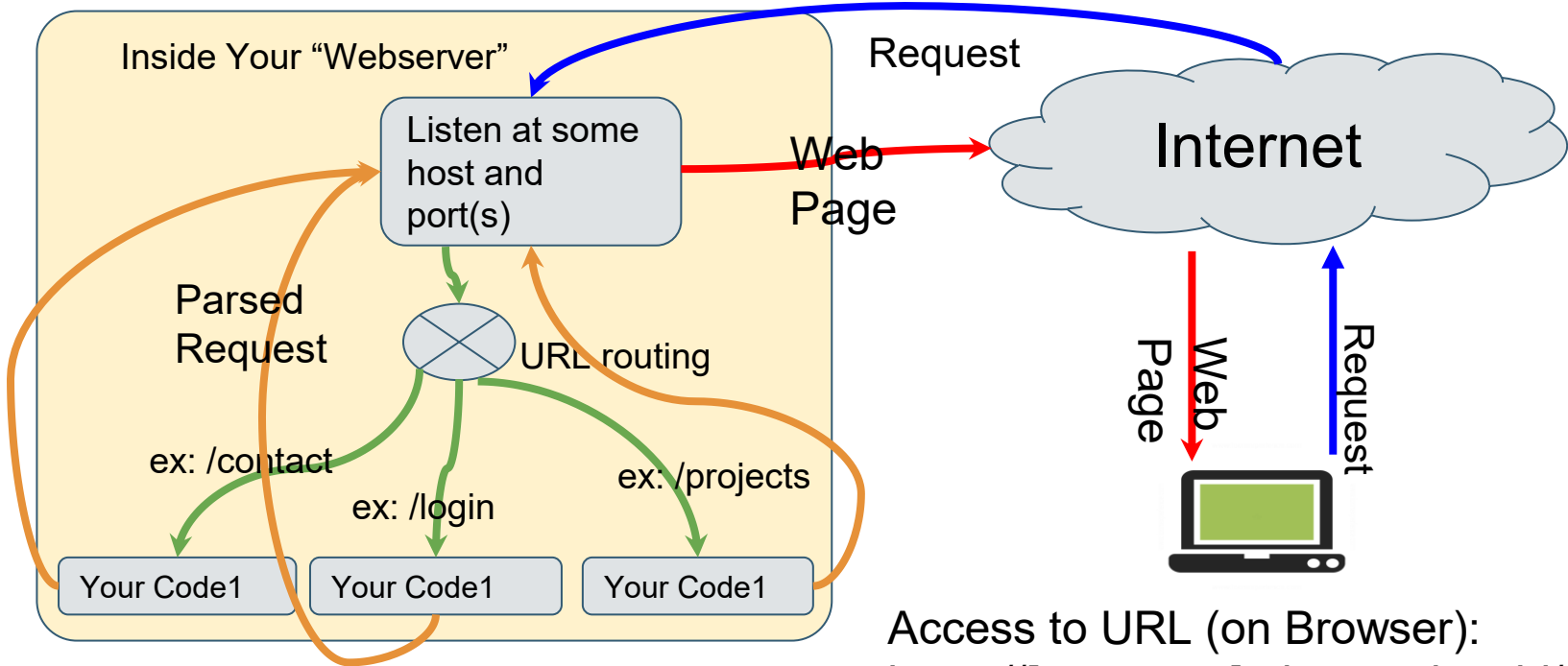
1. <https://en.example.com/>
2. <https://example.com/en/>

1. <https://www.youtube.com/watch?v=PH2QZHv4Ocw>
2. <https://youtu.be/PH2QZHv4Ocw>
3. <https://www.youtube.com/watch?v=PH2QZHv4Ocw&feature=youtu.be>
4. https://www.youtube.com/playlist?list=PLxuSBzNBcQcqLyLA_6ZaY1Rlr0QgdpPOg
5. https://www.youtube.com/watch?v=PH2QZHv4Ocw&list=PLxuSBzNBcQcqLyLA_6ZaY1Rlr0QgdpPOg&index=2

Web Request / Response



How Webserver Works



Access to URL (on Browser):
[https://\[appname\].pbp.cs.ui.ac.id/](https://[appname].pbp.cs.ui.ac.id/)

Web Programming

Web Programming

The Internet and
web programming technologies

To design web pages and applications
that run across an enormous range of
Internet-enabled devices.

- **Client-side programming** technologies are used to build web pages and applications that are run on the client (i.e., in the browser on the user's device).
- **Server-side programming**—the applications that respond to requests from client-side web browsers, such as searching the Internet, checking your bank-account balance, ordering a book from Amazon, bidding on an eBay auction and ordering concert tickets.

Client vs Server Side Scripting

CLIENT SIDE SCRIPTING	SERVER SIDE SCRIPTING
- Runs on the user's computer (Browser or scripting host)	- Executed on the web server
- Depends on browser → Browser must support the scripting language	- Not depend on client: Wide Range Programmatic capabilities→ Programmers have more flexibility
- User can view the script by using the browser's source viewing capability	- User cannot view the source code because the response is an HTML Page

Server Side Scripting Language



Client Side Scripting Languages

HTML



CSS



JS



Review

A programming framework is

a **prepackaged set of solutions** that solves common development problems.

Web Framework

Web Framework

- A web framework is a set of libraries and useful tools that can be used to build dynamic web applications.
- A web application framework forms generic web-based software application that is extended and modified by developers to create a specific application.

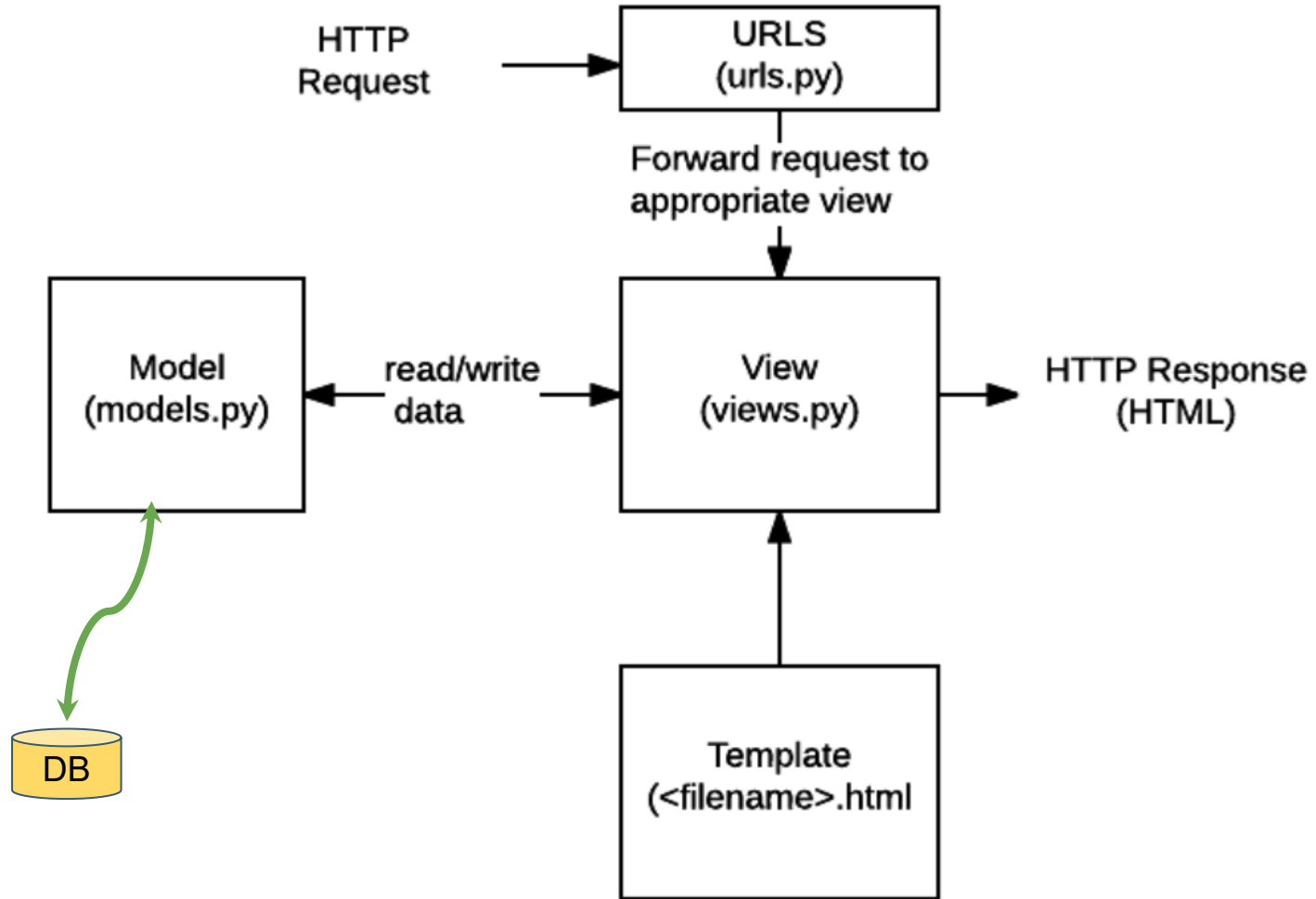
Why we use web frameworks

- Efficiency: provide basic building blocks
- Code reusing
- Separation of concerns
- Community support
- Structure matters

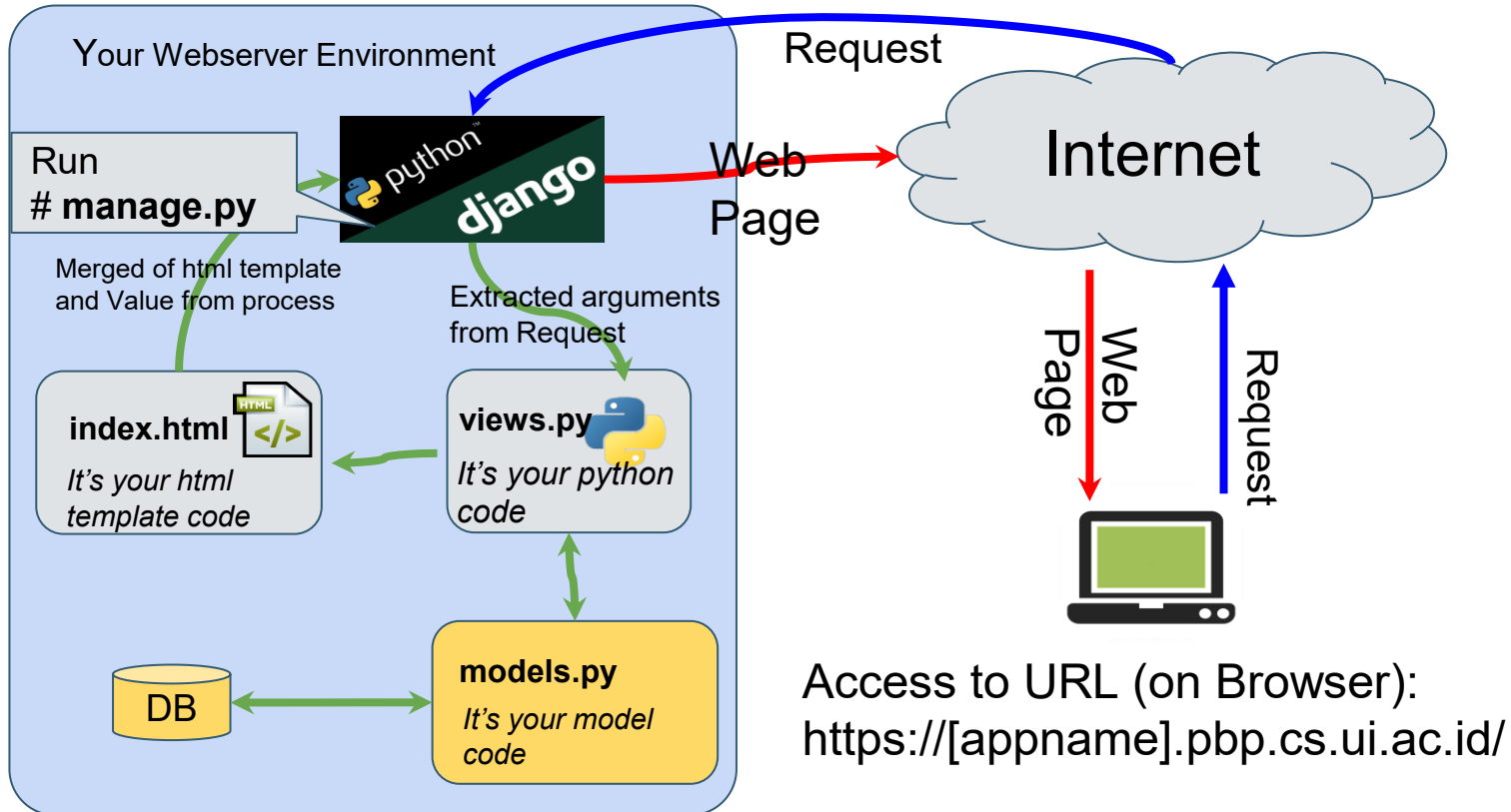
Get Stuff Done!

Django

- <https://www.djangoproject.com/>
- Language programming: Python
- Architectural pattern: Model–Template–Views (MTV)
- Why we use Django?
 - ☐ Open source
 - ☐ Ridiculously fast
 - ☐ Fully loaded
 - ☐ Reassuringly secure
 - ☐ Exceedingly scalable
 - ☐ Incredibly versatile



How Django Framework Works



Other Frameworks

Spring

- <https://spring.io/projects/spring-boot>
- Language programming: Java
- Architectural pattern: Model-View-Controller (MVC)

Laravel

- <https://laravel.com/>
- Language programming: PHP
- Architectural pattern: Model-View-Controller (MVC)

Yii

- <https://www.yiiframework.com/>
- Language programming: PHP
- Architectural pattern: Model-View-Controller (MVC)

Common Features of Web Framework

- url routing
- views
- database interaction
- form controls
- user management
- session management

A programming framework is a **prepackaged set of solutions** that solves common development problems.

//what else?

Summary

In this session, students have learned about:

- The basic concepts of the Internet
- How webserver works
- Web framework

Next Session Topics

- Django architecture
- TDD (Test Driven Development)

References

- Deitel: Internet & World Wide Web How to Program 4th Edition, Prentice Hall, 2008.
- Deitel: Internet & World Wide Web How to Program 5th Edition, Prentice Hall, 2011.
- "What is the difference between the Web and the Internet?", accessed on <https://www.w3.org/Help/#webinternet> (July 2024).
- "Syntax Components", accessed on <https://datatracker.ietf.org/doc/html/rfc3986#section-3> (July 2024).